Math 54 — Week 1

Solutions

P1.

Idea. Clear the first column with row 1, then the second column with row 2.

$$\left[\begin{array}{ccc|c} 1 & 1 & 2 & 9 \\ 2 & 4 & -3 & 1 \\ 3 & 6 & -5 & 0 \end{array}\right] \xrightarrow[R_3-3R_1]{R_2-2R_1} \left[\begin{array}{ccc|c} 1 & 1 & 2 & 9 \\ 0 & 2 & -7 & -17 \\ 0 & 3 & -11 & -27 \end{array}\right] \xrightarrow{2R_3-3R_2} \left[\begin{array}{ccc|c} 1 & 1 & 2 & 9 \\ 0 & 2 & -7 & -17 \\ 0 & 0 & -1 & -3 \end{array}\right]$$

1. $-x_3 = -3$, so $x_3 = 3$. Then $2x_2 = -17 + 21 = 4$, so $x_2 = 2$. Then $x_1 = 9 - 2 - 6 = 1$.
2. Check: $1 + 2 + 6 = 9$; $2 + 8 - 9 = 1$; $3 + 12 - 15 = 0$. Solution $(1, 2, 3)$.

(The step $2R_3 - 3R_2$ scales row 3 by 2 and then adds -3 times row 2; doing the scaling first avoids fractions.)

P2.

Idea. Inconsistent exactly when a row becomes $[\,0\ 0\ b\,]$ with $b \neq 0$.

$R_2 + 2R_1$ gives

$$\left[\begin{array}{cc|c} 1 & -3 & h \\ 0 & 0 & 2h-5 \end{array}\right].$$

The second row is $0 = 2h - 5$. It is consistent exactly when $2h - 5 = 0$, i.e. $h = 5/2$.

P3.

Idea. The second pivot is $8 - 4h$, which may be zero; split into cases.

$R_2 - 4R_1$ gives

$$\left[\begin{array}{cc|c} 1 & h & 2 \\ 0 & 8-4h & k-8 \end{array}\right].$$

1. If $h \neq 2$: $8 - 4h \neq 0$ is a pivot, both variables are basic, and the rightmost column is not a pivot column. Exactly one solution (for every k).
2. If $h = 2$: the second row is $[\,0\ \ 0\ \ k-8\,]$. If $k \neq 8$ it says $0 = k - 8 \neq 0$: no solution. If $k = 8$ it is a zero row; x_2 is free: infinitely many solutions.

(a) $h = 2$, $k \neq 8$. (b) $h \neq 2$. (c) $h = 2$, $k = 8$.

P4.

Idea. Forward phase, then backward phase from the rightmost pivot.

1. $R_3 - 2R_1$: row 3 becomes $[\,0\ 0\ 2\ -8\ 2\ \ 5\,]$. Then $R_3 - 2R_2$: $[\,0\ 0\ 0\ 0\ -2\ \ -5\,]$. Echelon form; pivots in columns 1, 3, 5.
2. Scale row 3 by $-\frac{1}{2}$: $[\,0\ 0\ 0\ 0\ 1\ \ \frac{5}{2}\,]$. Clear above it: $R_2 - 2R_3$ gives $[\,0\ 0\ 1\ -4\ 0\ \ 0\,]$.
3. Clear above the pivot in column 3: $R_1 + R_2$ gives $[\,1\ -2\ 0\ -1\ 0\ \ -2\,]$.

$$\text{RREF: } \left[\begin{array}{ccccc|c} 1 & -2 & 0 & -1 & 0 & -2 \\ 0 & 0 & 1 & -4 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & \frac{5}{2} \end{array}\right]$$

Basic: x_1, x_3, x_5. Free: x_2, x_4. Solution: $x_1 = -2 + 2x_2 + x_4$, x_2 free, $x_3 = 4x_4$, x_4 free, $x_5 = \frac{5}{2}$.

P5.

1. (a) False. The coefficient matrix has 3 rows, so at most 3 pivots, but 4 variable columns: at least one free variable. If consistent, infinitely many solutions.
2. (b) False.
$$\left[\begin{array}{cc|c} 1 & 0 & 1 \\ 0 & 1 & 2 \\ 0 & 0 & 0 \end{array}\right]$$
has a zero row and exactly one solution. Count free variables instead.
3. (c) True. Four pivots in a 4×4 coefficient matrix means a pivot in every row. Then the augmented column cannot contain a pivot (each row's leading entry is already in a variable column), so the system is consistent; and there are no free variables, so the solution is unique.

P6.

Idea. Each point gives one linear equation in a_0, a_1, a_2.

$p(1) = 4$: $a_0 + a_1 + a_2 = 4$. $p(2) = 9$: $a_0 + 2a_1 + 4a_2 = 9$. $p(3) = 18$: $a_0 + 3a_1 + 9a_2 = 18$.

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 4 \\ 1 & 2 & 4 & 9 \\ 1 & 3 & 9 & 18 \end{array}\right] \xrightarrow[R_3 - R_1]{R_2 - R_1} \left[\begin{array}{ccc|c} 1 & 1 & 1 & 4 \\ 0 & 1 & 3 & 5 \\ 0 & 2 & 8 & 14 \end{array}\right] \xrightarrow{R_3 - 2R_2} \left[\begin{array}{ccc|c} 1 & 1 & 1 & 4 \\ 0 & 1 & 3 & 5 \\ 0 & 0 & 2 & 4 \end{array}\right]$$

$a_2 = 2$, $a_1 = 5 - 6 = -1$, $a_0 = 4 + 1 - 2 = 3$. So $p(t) = 3 - t + 2t^2$. Check: $p(1) = 4$, $p(2) = 9$, $p(3) = 18$.

P7.

Idea. Four equations in two unknowns usually have no solution; row reduce to see.

$$\left[\begin{array}{cc|c} 1 & 2 & 5 \\ 3 & -1 & 1 \\ 2 & 3 & 8 \\ 1 & 1 & 3 \end{array}\right] \to \left[\begin{array}{cc|c} 1 & 2 & 5 \\ 0 & -7 & -14 \\ 0 & -1 & -2 \\ 0 & -1 & -2 \end{array}\right] \to \left[\begin{array}{cc|c} 1 & 2 & 5 \\ 0 & 1 & 2 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{array}\right]$$

(Second step: scale row 2 by $-\frac{1}{7}$, then add it to rows 3 and 4.) No row $[\,0\ 0\ b\,]$ with $b \neq 0$: consistent, no free variables. $y = 2$, $x = 1$. Check all four: 5, 1, 8, 3.